

(PROJECT PROPOSAL)



Temporary Rental Space Management System of MFU

**PANITI KONNGOEN
PRATHANAKORN PITI**

**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2026

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY
Temporary Rental Space Management System of MFU

**PANITI KONNGOEN
PRATHANAKORN PITI**

**A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2026

**© COPYRIGHT BY MAE FAH LUANG UNIVERSITY
Temporary Rental Space Management System of MFU**

**PANITI KONNGOEN
PRATHANAKORN PITI**

THIS COMPUTER ENGINEERING PROJECT HAS BEEN APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING
2026

EXAMINING COMMITTEE

..... ADVISOR

(Dr. Surapong Uttama)

..... COMMITTEE

(Dr. Paweena Suebsombut)

..... COMMITTEE

(Dr. Khwunta Kirimasthong)

ACKNOWLEDGEMENT

This project was completed with the valuable support and guidance of many individuals. The authors would like to express their sincere gratitude to the advisor and supervisory committee members for their professional guidance, constructive suggestions, and continuous support throughout the project.

The authors also wish to thank Mae Fah Luang University for providing the facilities and resources necessary for this study. Finally, heartfelt appreciation is extended to the authors' parents and friends for their encouragement and support during the completion of this project.

Paniti Konngoen

Prathanakorn Piti

Title	Temporary Rental Space Management System of MFU	
Author	Mr. Paniti Konngoen	
	Mr. Prathanakorn Piti	
Degree	Bachelor of Engineering	
	(Computer Engineering)	
Supervisory Committee	Dr. Surapong Uttama	Advisor
	Dr. Paweena Suebsombut	Committee
	Dr. Khwunta Kirimasthong	Committee

Abstract

This senior project proposes the development of a web-based room reservation system for Mae Fah Luang University. The purpose of this system is to improve the efficiency and convenience of reserving university rooms by providing an online platform for students, lecturers, and staff members.

The system allows users to view room availability, make reservation requests, manage booking schedules, and check reservation status through a user-friendly website. Administrators can manage rooms, approve or reject reservation requests, and monitor system usage.

The developed system aims to reduce manual processes, prevent scheduling conflicts, and enhance overall room management efficiency. System testing results indicate that the website operates according to the defined requirements and can effectively support room reservation activities within the university.

Keyword: Room Reservation System, Web Application, University Management, Online Booking, Mae Fah Luang University

CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Room reservation management is an essential service in educational institutions, especially in universities where classrooms, meeting rooms, and activity spaces are shared among many users. Traditional reservation methods, such as manual booking or paper-based systems, often lead to scheduling conflicts, inefficient communication, and difficulty in tracking room usage.

With the advancement of web technologies, online reservation systems have become an effective solution for managing shared resources. A web-based room reservation system can provide real-time availability information, reduce administrative workload, and improve accessibility for users.

Therefore, this project focuses on developing a room reservation website for Mae Fah Luang University to enhance the efficiency, accuracy, and transparency of room booking processes.

1.2 Objective

- 1.2.1 To develop a web-based room reservation system for Mae Fah Luang University
- 1.2.2 To enable users to view room availability and submit reservation requests online
- 1.2.3 To provide administrators with tools to manage rooms and reservation requests efficiently

1.3 Scope

- 1.3.1 General user
 - Check available dates and times and make reservations in advance within the specified time.
 - Reservation history.
 - Reservation purpose.
 - Status notification.
 - Cancel reservations.
- 1.3.1.1) Internal User (Within MFU Organization)
- 1.3.1.2) External User (Outside MFU Organization)
- 1.3.1.3) Collaborative Users (Cross-organization)

1.3.2 Administrator

- Manage rooms (Add/Disable/Edit)
- Update status (Approve/Disapprove)
- Manage Approval Workflows
 - o Process official approval documents and attach executive authorization files.
- Check availability date and time

1.4 Methodology

This project follows an iterative Software Development Life Cycle (SDLC) to ensure the system meets the requirements of Mae Fah Luang University.

- **1.4.1 Requirement Gathering and Analysis:** Collect and analyze user requirements by reviewing existing manual reservation processes and examining documents from the Intellectual Property and Innovation Management Division.
- **1.4.2 System Design:** Design the system architecture, user interface (UI), and database. This includes designing the relational database schema and mapping data relationships using PostgreSQL and pgAdmin 4.
- **1.4.3 System Development:** Implement the web application using a modern tech stack. The frontend will be developed using Vue.js and Tailwind CSS for a responsive user interface, while the backend API and server logic will be handled by Node.js.
- **1.4.4 System Testing and Evaluation:** Conduct rigorous testing to identify and resolve bugs. This includes testing role-based access for internal and external users, validating the approval workflows, and ensuring the system prevents scheduling conflicts.
- **1.4.5 Documentation and Presentation:** Prepare the formal Senior Project 1 and 2 documentation and defend the project before the examination committee.

1.5 Plan

Table 1.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 examination										
System development										
System testing and evaluation										
Senior project 2 document preparation										
Senior Project 2 examination										

1.6 Expected Result

- 1.6.1 A web-based room reservation system for Mae Fah Luang University
- 1.6.2 Users will be able to reserve rooms conveniently through an online platform
- 1.6.3 Improved efficiency and accuracy in room reservation management

1.7 Place and Equipment

1.7.1 Equipment

1.7.1.1 Hardware

- Personal computer or laptop
- Internet connection

1.7.1.2 Software

- Frontend: Vue.js, Tailwind CSS
- Backend: Node.js
- Database: PostgreSQL, pgAdmin4

1.7.1.3 Data set

- University room details and specifications